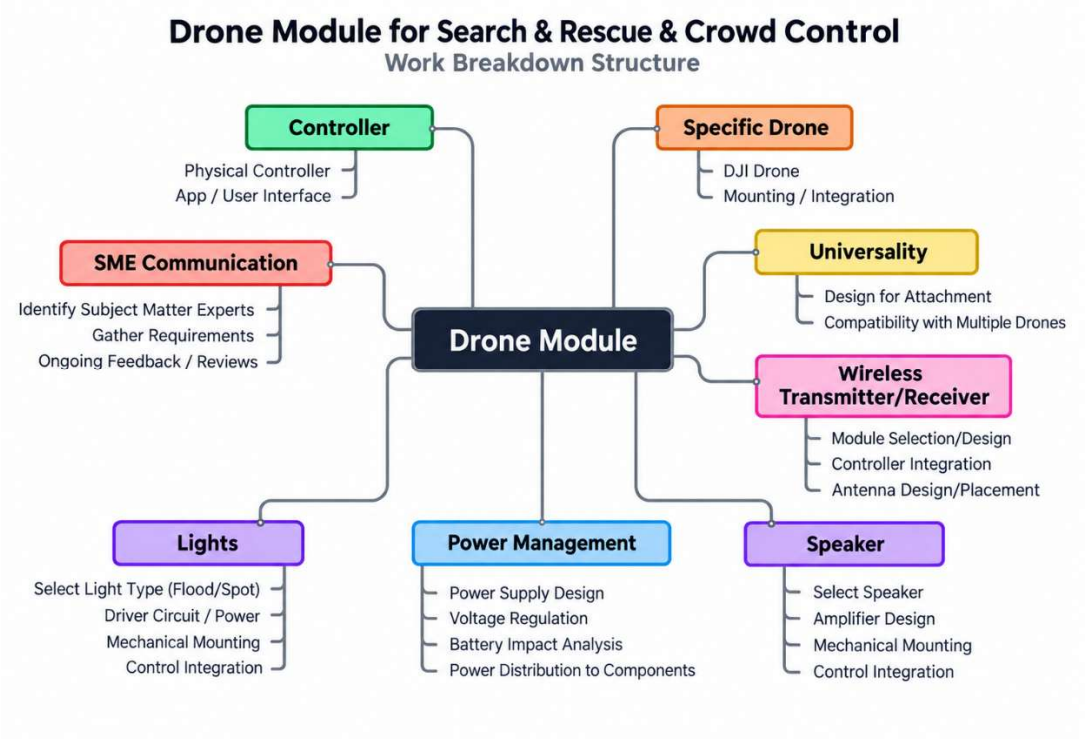
ELEC 4010 Capstone 1 Weekly Status Report

Project Name:	Drone Mounted Emergency Communication System
Team Members:	Hamilton Crews, Beau Blackburn, Tyler Engle, Joseph Rusk, Alejandro Matlacuatzi, Creed Allen
Week:	2
Date:	9/11/2026
Website link:	https://josephrusk.github.io/team2-capstone-site/

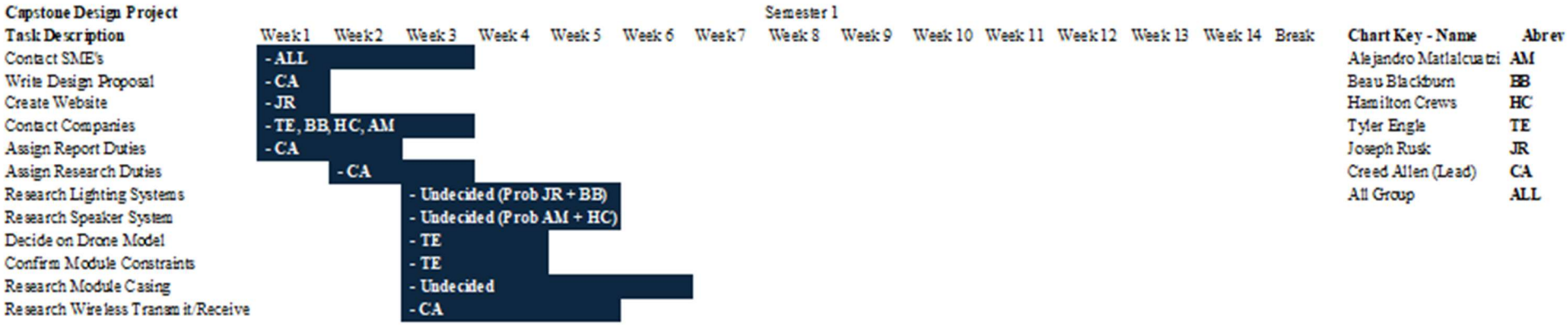
Project description in plain language, approximately 25 words:
Design and development of a drone mounted communications module hosting speakers, cameras, and lights. Module is to be used by search and rescue and law enforcement agencies to reach people that are unable to communicate through other means.

TASK STATUS - VISUAL

(1) Draw the Work Breakdown Structure for your project



(2) Show the updated Gantt Chart



Accomplishments since last status report: Decided to focus our primary project application towards search and rescue. Main subsystem focus will now be on high power, efficient LED floodlights, and a long-range speaker for locating and communicating with people that could otherwise not be. Applications for crowds and environments where sound is highly attenuated have been removed from the priority. Project extensions, in the case of early completion, include cameras, ideally with thermal imaging capabilities, and multi-module speaker synchronization.

Obstacles encountered since last status report: Have not gotten any confirmed subject matter experts to assist with our project. Possibility of exchanging our project for Dr. Lian's was discussed and deemed not desirable by members of the group.

Risks facing the project: Identifying subsystem components (lights+speaker) with sufficient efficiency and weight to be carried by a commercial off-the-shelf drone model. Battery power for the module will also take a heavy toll on the drones operating time and range.

Objectives for the next week: Confirmation of assistance from subject matter experts is the main focus with further project research coming afterwards.

Comments: Team worked well without an in-person meeting, bodes well for productivity throughout the projects development.